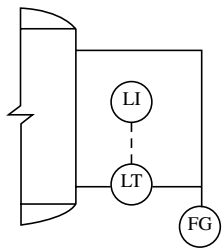
3.6 Differential Pressure Level Detectors

B. G. LIPTÁK (1969) **D. S. KAYSER** (1982) **J. A. NAY** (1995, 2003)



Flow Sheet Symbol

Design Pressure	To 10,000 PSIG (69 MPa)
Design Temperature	To 350°F (175°C) for differential-pressure (d/p) cell and to 1200°F (650°C) for filled systems; others to 200°F (93°C); standard electronics generally limited to 140°F (60°C)
Range	Differential-pressure cells and indicators are available with full-scale ranges as low as 0 to 5 in. (0 to 12 cm) H_2O. The higher ranges are limited only by physical tank size, since d/p cells are available with ranges over 433 ft H_2O (7 MPa or 134 m H_2O).
Inaccuracy	±0.5% to 2% of full scale for indicators and switches. For d/p transmitters, the basic error is from ±0.1% to 0.5% of the actual span. Added to this are the temperature and pressure effects on the span and zero. In case of "intelligent" transmitters, the pressure and temperature correction is automatic, and the overall error is ±0.1% to 0.2% of span with analog outputs and may approach ±0.01% with digital outputs.
Materials of Construction	Plastics, brass, steel, stainless steel, Monel®, and special alloys for the wetted parts. Enclosures and housings are available in aluminum, steel, stainless steel, and fiberglass composites, with aluminum and fiberglass being most readily available.
Cost	$200 to $1500 for transmitters in standard construction; $50 to $500 for local indicators. Add $400 to $1000 each for extended diaphragms and up to $1000 for "smart" features such as communications and digital calibration, although many "smart" features may be included in the base price. "Expert" tank systems cost approximately $1500 for each basic transmitter plus $3500 to $4500 each for one or more interface units and $1500 to $4000 for software plus a hand-held communicator and/or permanent connection to an in-house network. Some "expert" functions may be incorporated in an in-house network.
Partial List of Suppliers	See also Sections 5.6 and 5.7. Any d/p cell may be installed and connected to measure level if it has the appropriate accuracy and rangeability. Similarly, for vented tanks, any suitable gauge pressure device may be adapted to measure level. ABB Instrumentation Inc. (www.abb.com) Barton (www.ittbarton.com) Endress+Hauser Systems & Gauging (www.systems.endress.com) Enraf (www.enraf.com) Honeywell Control Products (www.honeywell.com/acs/cp/index.jsp) Rosemount Inc. (www.rosemount.com) Schlumberger Measurement Div. (www.slb.com/rms/measurement) Smar International Corp. (www.smar.com/products/function.asp) Viatran Corp (www.viatran.com) Yokogawa Corporation of America (www.yca.com)

For a general treatment of differential pressure devices, refer to Section 5.6. For a discussion of electronic pressure and differential pressure instruments, refer to Section 5.7.

Liquid level can be measured (inferred) by measuring a differential pressure (d/p) caused by the weight of the fluid column in a vessel balanced against a reference. For vessels at atmospheric pressure, the high side of the instrument is connected to the bottom of the vessel, and the low side (reference) is vented to the atmosphere. This method of level measurement is often referred to as *hydrostatic tank gauging*

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(HTG), especially in the bulk liquid industries. For pressurized vessels, the reference side must also be connected to the vessel so that both sides of the instrument equally sense the static pressure changes within the vessel and their differential responds only to the d/p caused by fluid head. The reference can be a column of fluid of fixed height such as a liquid-filled reference leg outside the tank or, as in the case of a bubbler system, a gas-filled reference leg, usually inside the tank. The key requirement is that the reference leg provide (or represent) a constant, known, hydrostatic head. An increase in the tank level from empty (0% level) to full (100% level) can result in a readout of 0 to 100% d/p (direct acting) or in a readout of 100 to 0% d/p (reverse acting). To obtain an accurate measurement using d/p cells, the densities of the process liquid and of the reference leg must be known and either be constant or continuously considered.

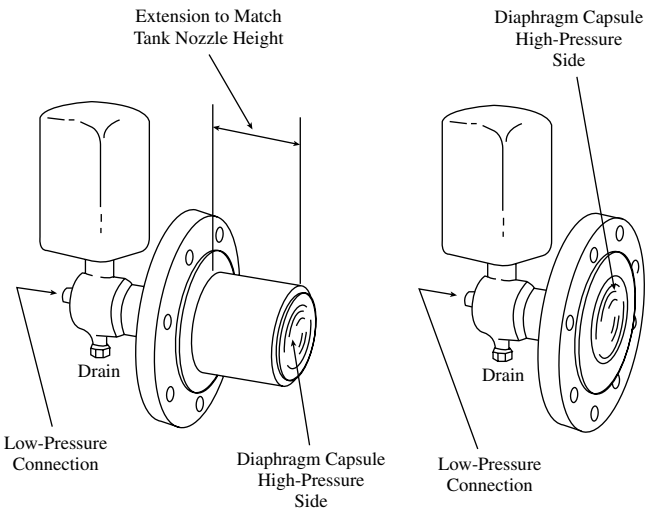


FIG. 3.6a
Extended diaphragm d/p cell (left); flat diaphragm d/p cell.

SENSING DIFFERENTIAL PRESSURE

Differential pressure can be detected by sensing two pressures separately and taking the difference to obtain liquid level. In practice, however, it is generally desirable to use a single pressure-difference sensor so that the static pressure levels are balanced before any measurement errors are introduced. The importance of this consideration can be visualized, for example, on a 0- to 100-in. (0- to 2.5-m) water column (WC) measurement where the expected accuracy is ±0.5 in. (±13 mm) WC. It would be virtually impossible to approach this accuracy if the measurements were made at a static pressure of 1000 PSIG (7 MPa) using two independent sensors. Since 1000 PSI (7 MPa) is equivalent to about 28,000 in. (714,000 mm) WC, the required accuracy of each measurement would be ±0.25 in./28,000 in., or 0.0009% (7 mm/714,000 mm). Nevertheless, several highly accurate HTGs do use two or more transmitters for gauging tanks at atmospheric pressures (Figure 3.6e). For instance, a third transmitter located a fixed distance above the bottom transmitter can be used to sense, in real time, the difference in pressure caused by the density of the fluid. Its sensing line to the tank must always be submerged and full to ensure accurate readings. However, when the tank is being initially filled or completely emptied, a temperature measurement in the fluid may be used to calculate an estimate of density or to help extrapolate from earlier density measurement records.

Extended Diaphragms

The d/p cell can be modified for use on viscous, slurry, or other plugging applications. Figure 3.6a shows two such designs: the extended and the flat diaphragm d/p cell transmitters. Their principles of operation are the same as those of the conventional d/p cells, except that the high-pressure side of the diaphragm capsule is exposed and extended to be in direct contact with the process. The extended diaphragm version is designed to bolt directly to the vessel nozzle; the protrusion can be sized to fill the space in the nozzle, placing the diaphragm flush with or slightly inside of the vessel wall. This design completely eliminates dead-ended cavities and is used especially on materials that can freeze at high temperatures or that can deteriorate or discolor if pocketed. This design cannot be serviced without depressurizing and draining the vessel.

The flat diaphragm design is normally installed by bolting it directly to a block valve on the vessel nozzle. A full-size gate or ball valve can allow for service without draining or depressurizing the tank. Because the connection is large, typically 3 in. (76 mm), the process is less apt to bridge or plug the sensing connection. Flat diaphragm cells can also be furnished with a solvent flush or steam-out connection. Both the flat and the extended designs are available with ranges up to 850 in. (2160 mm) with an accuracy of ±0.5% of span. The flat units can withstand 550 PSIG (3.8 MPa) operating pressures when the process is at 350°F (175°C). The maximum process temperature rating for the extended design is 750°F (400°C). Changes in process or ambient temperatures can cause zero shifts, as is the case with many similarly designed instruments. To minimize this effect, the d/p cell should be zeroed at the normal operating temperature, and the exposed body of the transmitter should be insulated. Alternatively, temperature at the connection or within the transmitter may be measured and continuously accounted for in the computation of level.

Some suppliers offer the extended and flat diaphragms with Teflon®, Viton®, or other plastic coating. This coating is intended as a slippery surface to minimize material buildup on the diaphragm. Do not rely on the plastic coatings for corrosion protection unless the supplier states specifically that the coating is so designed. As a general rule, engineers should not rely on coatings for corrosion protection of wetted parts of most process instruments, because, if the coating is nicked during installation, that protection is destroyed.

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Chemical Seals

Transmitters may also be provided with liquid-filled extension elements (chemical seals). The units shown in Figure 3.6a are used on atmospheric tanks or on pressurized tanks if the low-side connection can be kept clean or sealed. The designs shown in Figures 3.6b and 3.6c are used on pressurized vessels where plugging or corrosion can occur on both the high- and the low-pressure sides. The chemical seal designs are available in a broad range of materials including such metals as tantalum and zirconium. Vessel connection considerations apply to both the high and the low side. As can be seen from Figure 3.6c, the process material contacts both the diaphragm and the process side of the flange.

The instrument side of the diaphragm is filled with an oil or other suitable fluid (sometimes including water) and is connected by capillary to the high and low sides of the d/p cell. The differential pressure capabilities of these systems depend on the d/p cell selected. The accuracy of the system will always be worse than the accuracy of the d/p cell itself.

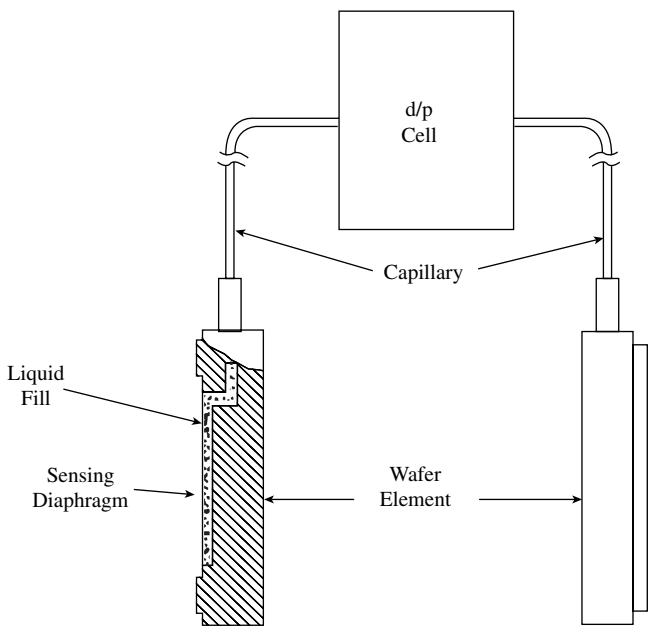


FIG. 3.6b
d/p cell with wafer elements.

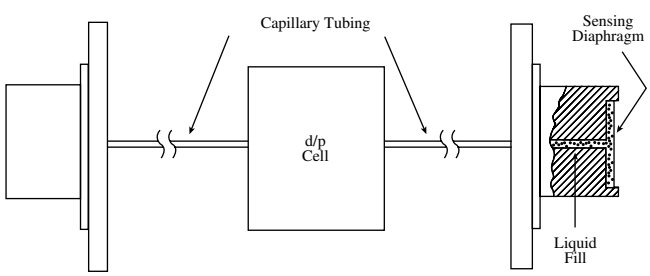


FIG. 3.6c
d/p cell with extended chemical seal elements.

The spring constant of the diaphragms at the chemical seals and the elasticity of the fill fluid will introduce some error, not all of which can be fully compensated. This error becomes more pronounced at very small differential pressure ranges and at high static pressures. A larger and less predictable error can result from the temperature-sensitive nature of the seal and capillary system. Temperature differences between the low and high sides will cause differing amounts of thermal expansion; this will be sensed by the d/p cell as a differential pressure and interpreted as a level change. Because the unequal amounts of expansion can be caused by the process temperatures or by changes in the ambient conditions, it is not always possible to zero-out this error.

The pressure and temperature ratings for these filled systems depend on seal design and filling liquid. Seals are readily available that are rated to 1500 PSIG (10 MPa). The fill material is normally a silicone oil that is good to 450°F (232°C). Other fill materials raise temperature ratings to 1200°F (650°C), allowing chemical seal designs to be considered for high-temperature applications. The volume of fluid in the seal, capillary, and d/p cell housing may be significant. Process, environmental, or health sensitivity to accidental leakage of the fill fluid may restrict choices. Distilled water has been successfully used as a fill fluid in some cases.

INTELLIGENT D/P CELLS AND TANK EXPERT SYSTEMS

The microprocessor has extended the applications of electronic differential-pressure transmitters. The detection is still accomplished by use of the elastic element (diaphragm or bellows), although the actual elastic element may be very small. The elastic element may actually have some of the electronic and microprocessor circuits attached to, embedded in, or otherwise incorporated directly. The microprocessor can convert the analog readings of deflection to a linearized, high-resolution digital signal compensated for temperature effects. Transmitter stability is thus improved, and the time between physical calibrations is often extended to more than a year.

"Smart" level transmitters can convert the level readings of spherical or cylindrical tanks into actual volume percentage readings (Figure 3.6d). Intelligent transmitters have also been

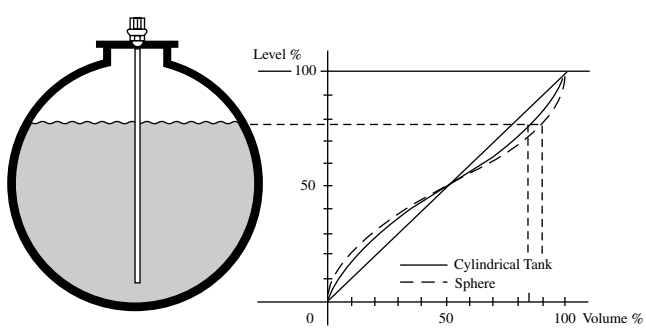


FIG. 3.6d
Intelligent transmitters can automatically convert level readings into volume.

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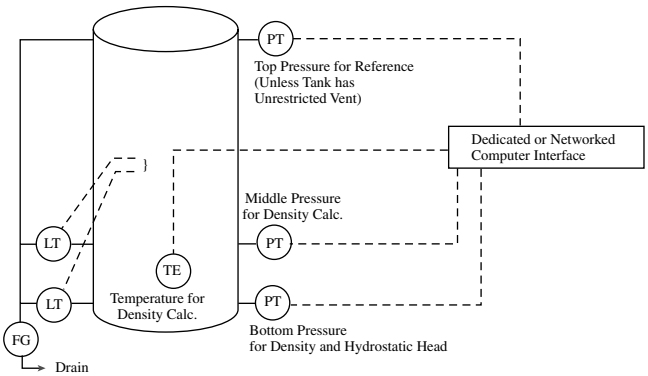


FIG. 3.6e
Multiple transmitters to provide level, mass, volume, and density data.

combined into tank expert packages that, in addition to level, can also calculate mass, density, and volume based on the measurements from three or more d/p cells and one temperature transmitter as shown in Figure 3.6e. Outputs are available through digital RS485 and RS232 connections and may be networked digitally for multiple remote access to the data. Most manufacturers offer optional digital communication with the transmitter through the output signal wires coincident with the analog output signal. Wireless Ethernet local area networking per IEEE-802b and other wireless technologies is also available. Calibration, range and zero setting, elevation, suppression, and linearization may all be accomplished remotely, although periodic physical calibration to a known NIST traceable standard is still recommended for the most accurate results.

In Figure 3.6e, the "TOP" pressure transmitter can be eliminated if the tank is completely vented without any possible vent restrictions. Simple problems, such as a bird's nest in a vent pipe, may cause unexpected and possibly undetected errors. This is a particular concern under transient conditions where the restricted vent does not allow enough flow to keep up with filling or draining operations. If the top transmitter is eliminated, it may be prudent to install a very low-pressure detector to alert operators of unexpected pressure or vacuum conditions. A better solution would be to use differential-pressure level transmitters (LTs) for the middle and bottom locations as shown on the left in Figure 3.6e. In this case, they should share a common dry reference leg.

The bottom and middle transmitters, whether pressure or differential pressure, must be placed a known vertical distance apart. Because horizontal distance does not matter, on large tanks, it would facilitate maintenance if each were placed within reach of stairways, ladders, or platforms existing at the appropriate levels. Density is determined by the difference in readings between the middle and bottom transmitters. For instance, if the transmitters were placed 50 in. (125 mm) apart, and if the tank contains distilled water at 68°F (20°C), the bottom transmitter should read exactly 50 in. (125 mm) of water more than the middle transmitter. If the tank contained an oil with a specific gravity of 0.75, then the difference in readings would be 37.5 in. (93.75 mm) of water. Anything that changes the density, including bulk temperature changes and chemical composition changes, will be reflected in the difference in readings between the middle and bottom transmitters. Thus, the density can always be calculated.

A temperature reading may be useful as a diverse measurement to verifying expected density or to assist in estimating or extrapolating density as the tank is initially filled or as it is emptied below the middle transmitter.

PRESSURE REPEATERS

When detecting the level in pressurized vessels, the vapor space pressure must be connected to the low side of the d/p cell to serve as a reference. On hard-to-handle materials, a one-to-one pressure repeater may be used to provide this reference and simultaneously isolate the d/p cell from the process. Repeaters develop an air output pressure equal to the vapor-space pressure. They are inexpensive, but their accuracy is limited. The error in the repeated output pressure increases as the repeated pressure rises. At a pressure level of 40 PSIG (0.27 MPa), the error is 2 in. (51 mm) H_2O; at a pressure level of 400 PSIG (2.7 MPa), the error is 20 in. (508 mm) H_2O. Obviously, errors of this magnitude are not acceptable for most process level measurements. Repeaters have generally fallen into disuse in favor of high-accuracy digital transmitters made from the same materials.

DRY, MOTION BALANCE DEVICES

These differential pressure detectors are also referred to as *bellows meters,* because they depend on liquid-filled, double-opposed bellows. Bellows meters are most useful where local indication or recording is required and where compressed air and electric power are not available as energy sources. They can be very sensitive to low differential pressures because of the large area and slack resistance to motion that can be built in.

Figure 3.6f illustrates the high- and low-pressure chambers, the range spring, and the drive assembly (a bell crank of sorts) to transfer bellows motion to the readout pointer. The bellows in both chambers and the passage between them are liquid filled. When the unit is installed, the pressure in the high-pressure chamber compresses the bellows so that the liquid flows from it into the low-side bellows. When the low-pressure (or range) bellows expands, it exerts a force against the range spring, which determines the span of the instrument. The linear motion of the range bellows moves the drive lever, mechanically transmitting a rotary motion through the sealed torque tube assembly to the indicator. The output motion from the torque tube assembly is limited to a few degrees of angular rotation. This is sufficient for most

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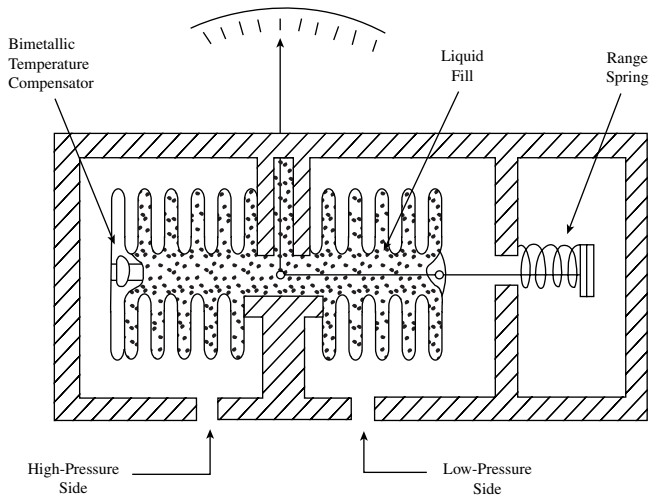


FIG. 3.6f
Motion detector d/p indicator.

mechanically driven local indicators or recorders and well within the capabilities of modern electronic motion sensors.

If the secondary device imposes a considerable load on the torque tube assembly, however, the accuracy and sensitivity of the unit can be destroyed. For sustained accuracy, bellows meters depend on the repeatability of their mechanical systems, which have proven to be linear within 0.5% to 1% of full range, and down to 0.2% or less when the range can be limited to a small portion of the total available motion. A temperature compensator built into the bellows assembly compensates for the changing volume of the fill liquid resulting from ambient temperature variations. Bellows meters are provided with overrange protection. The operation of one of the protection mechanisms is as follows: The bellows move in proportion to the differential pressure applied across them and in proportion to the spring rate of the bellows plus the resisting springs. When the bellows have moved over their calibrated travel, a valve mounted on the center stem seals against its seat, thereby trapping the fill liquid in the bellows. Because the liquid is essentially incompressible, the bellows are fully supported and will not rupture, regardless of the pressure applied. This overrange protection is furnished in both directions, protecting both bellows. Another design of overrange protectors involves the use of liquid-filled bellows with a number of diaphragm discs and spacer rings between them. As the bellows are subjected to overrange pressures, the diaphragms nest, and the metallic spacer rings form a solid stop, thereby fully protecting the bellows from rupture.

Bellows meters can detect full-range pressure differentials at least as low as 20 in. (508 mm) WC and as high as 400 PSIG (2.7 MPa). Measurement of low differentials is limited by the small forces available to actuate the motion detector mechanism. For very high differentials, the limitation is the mechanical strength of the bellows. Standard units are available with steel or stainless-steel housings and stainless-steel or beryllium copper bellows. For corrosive applications, other materials can be obtained or special, high-displacement-volume chemical seals can be used. Static pressure ratings up to 10,000 PSIG (69 MPa) are available as standard; operating temperature is limited to 200°F (93°C).

LIQUID MANOMETERS

These instruments are discussed in detail in Section 5.9. Their design variations include the U-tube, the well, and the float-type manometers. The float designs can provide remote readouts, whereas regular manometers can serve as the readout indicators for bubbler-type level sensors. Where the use of glass is not allowed, digital manometers using magnetic coupling between a float inside a stainless-steel U-tube manometer, and an outside electronic transmitting mechanism can be used.

Glass-tube manometers are available with ranges up to 120 in. (3.05 m), which is sufficient for many level applications. The magnetically coupled float manometers are available for high-pressure services, up to 6000 PSIG (41 MPa), and can measure up to 1000 in. (25 m) of water column.

Because of the fragile nature of glass, the toxicity of mercury, and the chemical interaction between the manometer filling fluids and the process, manometers are not widely used on process level measurement applications and are mostly restricted in their use to occasional utility services.

LEVEL APPLICATIONS OF D/P CELLS

The applications of pressure differential detectors as components in level-measurement loops will be covered in the following paragraphs. The requirements of atmospheric and pressurized tanks and the features of level loops on clean and hard-to-handle process fluids will be discussed separately. Figure 3.6g shows the symbols used for the

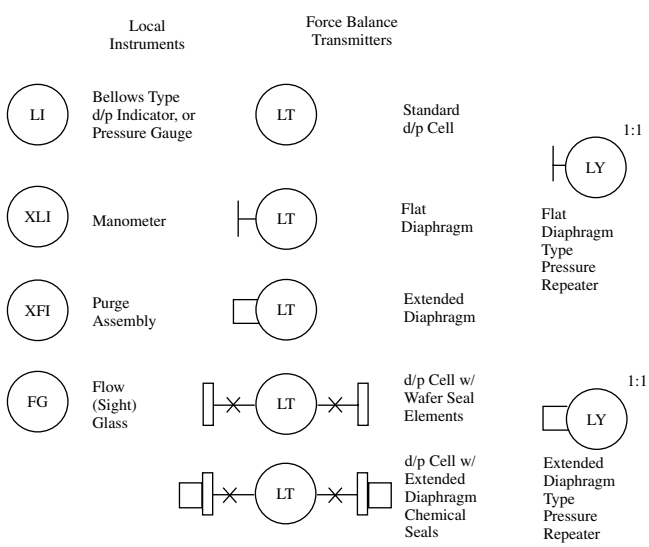


FIG. 3.6g
Symbols for d/p level loops.

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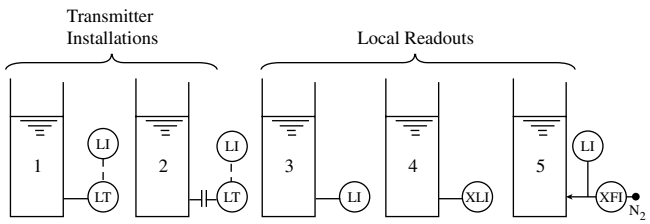


FIG. 3.6h
Detection of clean liquid levels in atmospheric tanks by d/p instruments.

various loop components. In the discussion that follows, the air-bubbler-type d/p cell installations will not be included, as these are covered in Section 3.2.

Clean Liquids in Atmospheric Tanks

Unpressurized vessels containing clean liquids are the least demanding as far as level measurement is concerned, because the two most common sources of difficulties (vapor pressure compensation and plugging) are not present. Figure 3.6h shows five tanks equipped with five different types of level devices. The first two are for remote readout, and the others are for local readout. On tank 1, a standard d/p transmitter with screwed connections is shown with its low-pressure side open to the atmosphere. This installation can be made by using a pressure transmitter instead of a d/p transmitter. The pneumatic receiver gauge is normally calibrated for 0 to 100% level. The flat diaphragm-type d/p transmitter is shown on tank 2. Compared to the standard d/p cell, the flat diaphragm-type transmitter is simpler to install, and it is nozzle mounted, requiring no other means of support. It also can be less expensive, because only the diaphragm and the retaining ring are in contact with the process. As a result, only these parts must be made of corrosion-resistant materials whereas, in the standard d/p cell, the entire body is exposed to the process fluids. A flat diaphragm-type pressure repeater can also be used in place of the d/p transmitter, in which case the receiver gauge will sense the actual hydraulic head instead of a 3 to 15 PSIG (21 to 103 kPa) transmitted signal. Tank 3 shows a motion balance local d/p indicator with the low-pressure side vented to atmosphere. The same measurement can be made by using a standard pressure gauge. The level in tank 4 is detected by a manometer. Although this is one of the most accurate and economical devices to use for local readout, the consequences of mechanical damage and proper selection of the filling liquid must be considered. The installation on tank 5 is basically an air bubbler system, which is detailed in Section 3.2. Not shown is a system (usually portable) wherein a miniaturized electronic transmitter is actually lowered, by a cable containing its wiring, into an atmospheric tank and down through the liquid until it reaches the bottom. The operator can tell that it is on the bottom when the level indication stops increasing. This is particularly useful for occasional tank gauging and for measuring the level in wells.

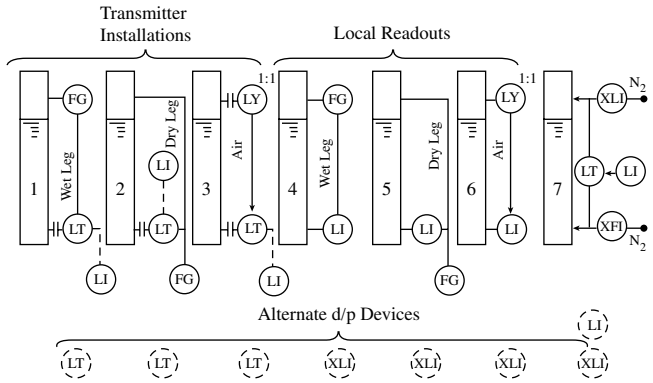


FIG. 3.6i
Measurement of clean liquid levels in pressurized tanks by d/p instruments.

Clean Liquids in Pressurized Tanks

When the level in a pressurized vessel is to be established by hydraulic head measurement, the instrument has to be compensated for the vapor pressure in the tank. This is done by exposing the low-pressure side of the d/p cell to these vapor pressures. Compensation can be achieved by various means. Figure 3.6i shows seven variations of this installation. Tank 1 illustrates a wet-leg application in which the compensating leg is prefilled with a chemically inert liquid that will not freeze or vaporize under operating temperature conditions. The wet-leg installations are used when the process vapors would otherwise condense into the compensating leg, thereby exposing the low-pressure side of the d/p cell to unpredictable hydraulic heads, or when the transmitter must be sealed from corrosive vapors. The prefilled wet leg creates a constant pressure on the low-pressure side of the transmitter. The leg is filled through a seal pot to provide excess capacity.

It is desirable to make this seal pot out of a sight flow indicator so that the level of the filling liquid is visible to the operator. The d/p cell can be either the standard or the flat diaphragm design. On tank 2, the same d/p transmitter is installed in a dry-leg system. This is acceptable when the process vapors are not corrosive and condensation at ambient temperatures is not expected. For such applications, a condensate pot is installed below the d/p cell, and it also should incorporate a sight flow indicator so that the operator can visually determine if it is time to drain out any accumulation of condensate. Tank 3 illustrates the use of a flat diaphragm pressure repeater for vapor pressure compensation. The problems associated with range depressor adjustments, corrosion, and condensate accumulation are eliminated by the use of repeaters, but they do add to the total error of the installation. On tanks 4, 5, and 6, the same basic installations (wet-leg, dry-leg, and repeater) are shown in connection with local bellows indicators. Manometers can also be considered in place of the motion balance d/p indicators if mechanical damage, chemical inertness of the filling fluid, and its compatibility with the operating temperatures are previously established. One concern with

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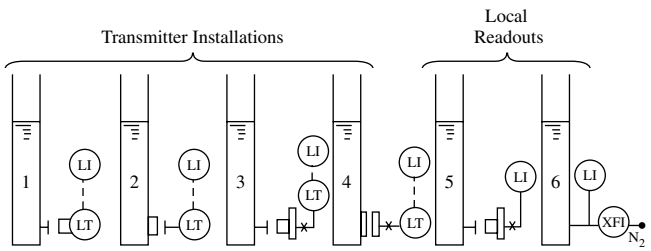


FIG. 3.6j
Sensing of hard-to-handle liquid levels in atmospheric tanks by d/p devices.

the dry-leg installation shown in tank 5 is the possibility that, if the vapors in the tank condense, or if the tank is flooded, process liquids will fill the dry leg. This is unsafe because, under these conditions, the d/p cell will signal a low level or an empty tank condition. The installation of a float trap can lower (but not eliminate) this risk by draining the dry leg if the liquid buildup is slow. Tank 7 illustrates a bubbler system with either transmitting or local indicating d/p devices. The advantages, limitations, and drawbacks of such installations have been pointed out in Section 3.2.

Hard-to-Handle Fluids in Atmospheric Tanks

Level measurement is more difficult when the process fluid is highly viscous, is likely to freeze, contains solids that can settle out, or can gel or polymerize in dead-ended cavities. Figure 3.6j shows six installations that may be considered for these conditions. On tank 1, an extended diaphragm-type force balance transmitter is shown. The diaphragm motion is limited to a few thousandths of an inch, and the nozzle cavity is completely filled by the diaphragm extension. The extended diaphragm transmitter is a good candidate for level measurement of hard-to-handle liquids, provided that the vessel can be drained when the transmitter needs service. Tank 2 is provided with a flat diaphragm transmitter mounted on a pad. The nozzle cavity is reduced but not eliminated. Tank 3 is furnished with liquid-filled chemical seals such as the one shown in Figure 3.6c. This unit will perform as well as the extended diaphragm d/p cell but, because of the liquid-filled capillary system, it is subject to errors caused by temperature variations. A wafer-type, liquid-filled element such as the one illustrated in Figure 3.6b is shown on tank 4. This sensing method combines the disadvantages of 2 and 3; the dead-ended cavity is not eliminated, and it is subject to temperature errors. However, it can be used on extremely hot processes. On tank 5, the element is the same extended chemical seal as on tank 3, but the readout is a local pressure gauge. An air bubbler is shown on tank 6.

Hard-to-Handle Fluids in Pressurized Tanks

When the process material in the vessel is hard to handle, it is frequently the case that the vapor space also contains materials such as foam that can build up and plug the sensing

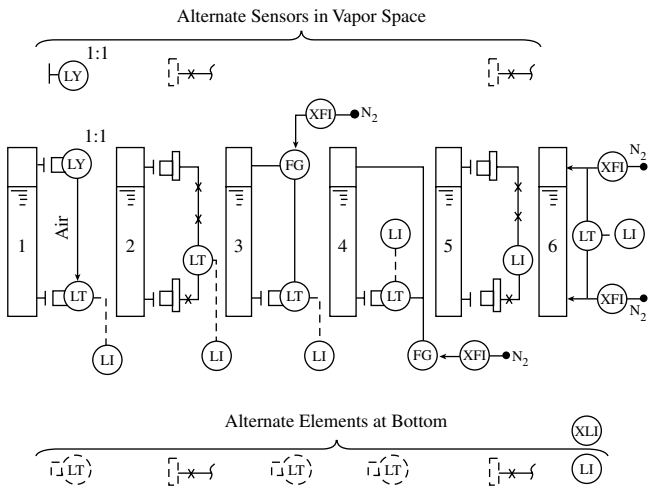


FIG. 3.6k
Detection of hard-to-handle liquid levels in pressurized tanks by d/p instruments.

line of the compensating leg. Figure 3.6k shows six methods for dealing with these applications. Tank 1 shows an extended d/p transmitter in the liquid region and an extended repeater in the vapor space. The use of these devices eliminates all possible plugging problems, because the sensing diaphragms are flush with the inside of the vessel wall. This detecting system will function properly on all except the most difficult crystallizer applications, where the inside wall of the tank might be coated with a layer of crystals.

The extended chemical seals shown on tank 2 will provide an installation similar to that on tank 1 and will perform similarly if temperature differences between the wafers or ambient temperature variations do not cause inaccuracies. Tanks 3 and 4 are equipped with extended d/p transmitters, but a purge flow prevents the process vapors from entering the compensating legs. The purge medium can be either liquid or gas and can be applied to both dry- and wet-leg installations. Such systems require additional maintenance and range depressor adjustments, and corrosion or condensate accumulation can cause calibration or reliability problems. Tank 5 shows a local indicator equipped with extended chemical seals. This device is subject to temperature effects and requires large displacement seals to match the displacement of the d/p indicator. For tanks 1 through 5, flat diaphragm elements can also be considered, but it should be realized that they do not completely eliminate the dead-ended nozzle cavities in which material can accumulate. Such designs should be considered only where it is essential to have an isolating valve between the tank and the level device so that it will not have to be drained prior to removal of the instrument. The bubbler system is illustrated on tank 6; either liquids or gases can be used as the purge media.

The extended diaphragm transmitter/repeater installation is attractive for pressure vessels containing hard-to-handle materials if isolation valves are not required. Purged installations are also acceptable if the availability of a purge media

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and maintenance are both reliable and if the process can tolerate accumulation of the purge flow.

SPECIAL INSTALLATIONS

One variation that can be considered is to install the d/p cell in a "reverse" arrangement wherein the high-pressure side is connected to the wet leg. Naturally, this can be done only with nonextended units such as the ones used on tanks 1 and 4 in Figure 3.6i. If this is done, these transmitters will detect the maximum d/p when the tank is empty and the minimum d/p when it is full.

On hard-to-handle processes where extended diaphragms are used, a "reverse-acting" d/p cell can be used when one or both sides are protected by chemical seals (such as in tanks 2 and 5 in Figure 3.6k) and the d/p is located near the bottom of the tank. In these cases, the high-pressure side of the d/p cell can be connected to the chemical seal element at the top of the tank. In that case, the maximum reading of the d/p cell occurs when the tank is empty.

Boiling Applications

Reverse-acting differential pressure level transmitters can be used on boiler or steam drum level applications (Figure 3.6l) where the wet leg is the high-pressure side of the process. When the process fluid condenses into a liquid at ambient temperatures, a wet-leg configuration can be obtained by allowing the condensate to accumulate in the wet leg rather than mechanically filling the wet leg with a slow drip. This can be achieved by installing an uninsulated condensate pot that remains at ambient temperature. Excess condensation from this pot drains back into the tank and therefore maintains a constant height of the reference wet leg.

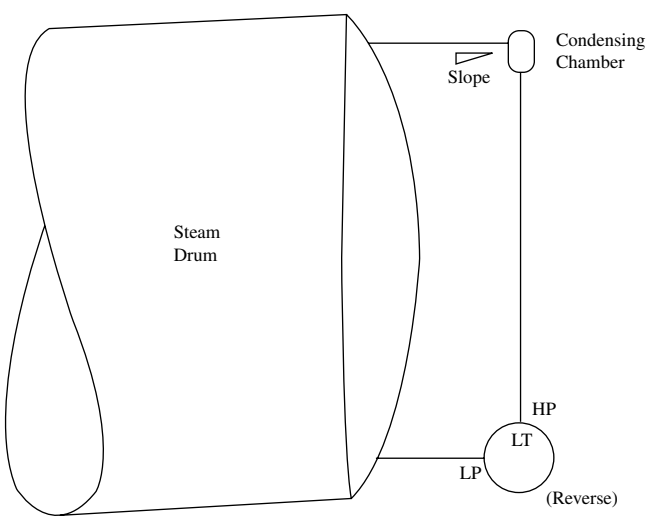


FIG. 3.6l
Level measurement in steam drums or on other boiling liquid applications.

One should understand that the output signal of such a d/p cell relates *not to the level inside the steam drum* but to the mass of water inside. If the condensate in the wet leg is cold, the wet-leg density will be substantially greater than that of the boiling fluid inside the drum. In addition, the density inside the drum will be a variable; density will drop as a result of the swelling effect when the steaming rate rises, and it will rise when the steaming rate drops. A level control system that adds colder feedwater to a steaming drum may result in bubble collapse such that the actual level decreases even further. The exact position of the *top surface* of the boiling fluid may never be determined. A large *void fraction* will result in a higher surface position without necessarily changing the differential pressure. A "full" tank or drum will not produce zero differential pressure under those conditions. Therefore, the d/p cell output can be converted into a true level reading only if the density and void fraction of the boiling fluid are separately accounted for using other available parameters, such as steaming rate, bulk fluid temperature, and/or discharge pressure (equivalent to temperature for saturated conditions). These are *very* important considerations when determining safety system setpoints. The shrinkage that may be caused by cold feedwater or a sudden reduction in steaming rate must be accounted for to avoid uncovering hot tubes. The swell that may be caused by an increasing demand transient must be accounted for to avoid liquid carryover. Nevertheless, differential pressure remains a popular method of measuring and controlling level in boiling vessels. Many safe and effective installations exist throughout industry, including many on nuclear and fossil-fueled boilers and steam generators in the power industry.

Cryogenic Applications

A similar situation exists when the process liquid is very cold, except that major bulk boiling rarely takes place within the vessel itself. The tank may be located inside high-thermal insulation, called a *cold box.* More often, it is located in a double-walled high-vacuum dewar tank (Figure 3.6m). These applications also involve boiling, but here the liquid nitrogen or other liquefied gas will, by design, boil primarily in external piping or heat exchangers. For sensing lines, this increase in temperature occurs as the liquid-filled pipe is approaching the wall of the cold box or nears the penetration point of the dewar. At some point in the sensing line, the liquid boils, causing a liquid–gas interface. From that point on, the sensing line is filled with vapor. To provide a stable and noise-free level signal, the installation should be such that the boiling will occur at a stable, well-defined point in the sensing line. Boiling should occur in a large-diameter, low-slope section of the sensing line as it approaches the penetration of the vessel outer wall. The line diameter should be large (1 in. or 25 mm), because the interface between the liquid and vapor can be turbulent during consumption transients. This low-slope section should approach the wall of the cold area so that the temperature will be high enough to guarantee

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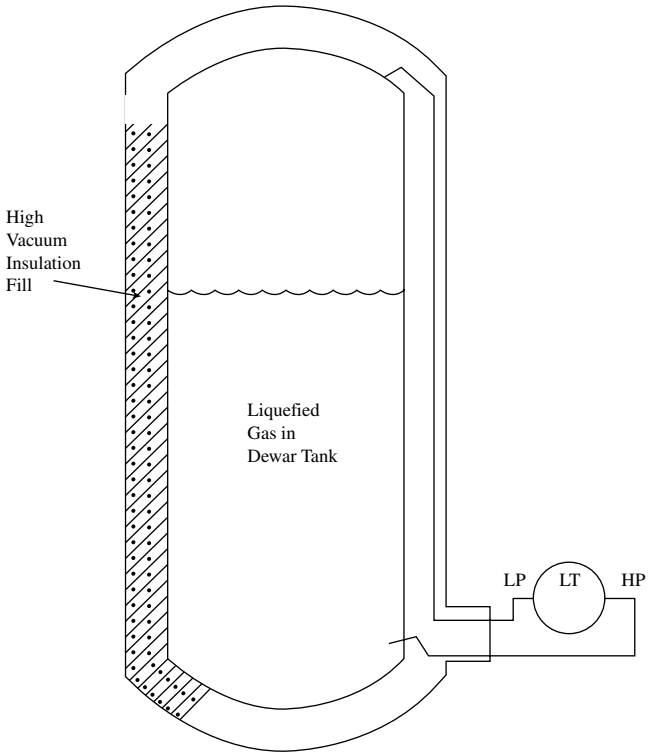


FIG. 3.6m
Cryogenic level measurement in vacuum-insulated tank.

that the process liquid is in the vapor form under all process and ambient conditions. To account for transient conditions that may temporarily move the interface point, a "dry" loop is often provided in the warmer portion of the sensing line so as to catch and quickly evaporate any temporary liquid carryover.

For most cryogenic level measurements, the density of gas in vertical sections of the sensing lines can safely be ignored. However, for argon (a very heavy gas), the effect should at least be calculated at limiting conditions to ensure that the weight will not exceed the desired accuracy of the measurement.

Normal Ambient Temperature Bi-phase Applications

Differential pressure may also be used for level measurement on large tanks containing gases such as refrigerants or propane under pressure, where a liquid–gas interface exists at normal ambient temperature. In these cases, a dry reference leg is suggested. A few watts of heat tracing along the dry reference leg will ensure that it remains a degree or two above ambient temperature (thus slightly superheated above saturation temperature) to avoid condensation of the gas. Care must be taken during fast filling operations to ensure that major condensation does not collect temporarily in the reference leg as a result of increasing pressure in the tank. Provision of a drip leg with a sight glass at the bottom of the reference leg is suggested.

TABLE 3.6n
D/P Cell Capsule Capabilities

	Low Range	*Medium Range*	*High Range*
Minimum span - in. H_2O	0–2	0–25	0–30 PSID
- kPa	0–0.5	0–6.2	0–210
Maximum span - in. H_2O	0–150	0–1000	0–3000 PSID
- kPa	0–37.5	0–250	0–207 bars
Maximum zero suppression	(maximum span) minus (calibrated span)		
Maximum zero elevation	Minimum span		

SPAN, ELEVATION, AND DEPRESSION

All d/p cells can be provided with zero, span, elevation, and depression adjustments, either mechanical or electronic. Table 3.6n shows some typical d/p cell ranges and the available elevation and suppression setting adjustments for each. Whenever the d/p is at an elevation other than the connecting nozzle on an atmospheric tank, the zero of the d/p cell needs to be elevated or depressed. It is important to realize that two zero reference points exist. One is the level in the tank that is considered to be zero (lower range value) when the tank is near empty. The other zero reference point is the point at which the d/p cell experiences a zero differential (zero value of the measured variable). The terms *elevation* and *depression* as used in this discussion refer to the zero experienced by the d/p cell (Figure 3.6o). This figure uses mechanical spring adjustments to physically illustrate the relationships. Equivalent electronic adjustments are available. In both cases, the overall available span must be wide enough to encompass the required adjustments. The tension in the elevation spring can be set to cancel out any initial pressure exerted on the high side of the diaphragm capsule. Similarly, the depression spring can be adjusted to compensate for initial forces on the low-pressure side of the d/p cell. The amount of depression setting is limited to the full range of the capsule, whereas the sum of elevation setting and span cannot exceed the full range of the cell. These settings are normally adjusted in the factory if sufficient data are furnished to the manufacturers. If the setting is changed in the field, it will affect the span of the transmitter.

Figure 3.6p shows a dry-leg d/p cell installation with the desired minimum and maximum liquid levels noted. The output of the transmitter will be zero when the level is at the minimum and 100% when it is at the predetermined maximum. The span (range) of the cell will be product of liquid density and the distance between minimum and maximum levels desired (X). The elevation spring will be set for the product of density times distance between the minimum level desired and the cell datum (Y). A reference leg is also shown on this sketch, which is convenient for checking the transmitter. Checking is done by temporarily isolating the cell from the tank and filling the reference leg with a known

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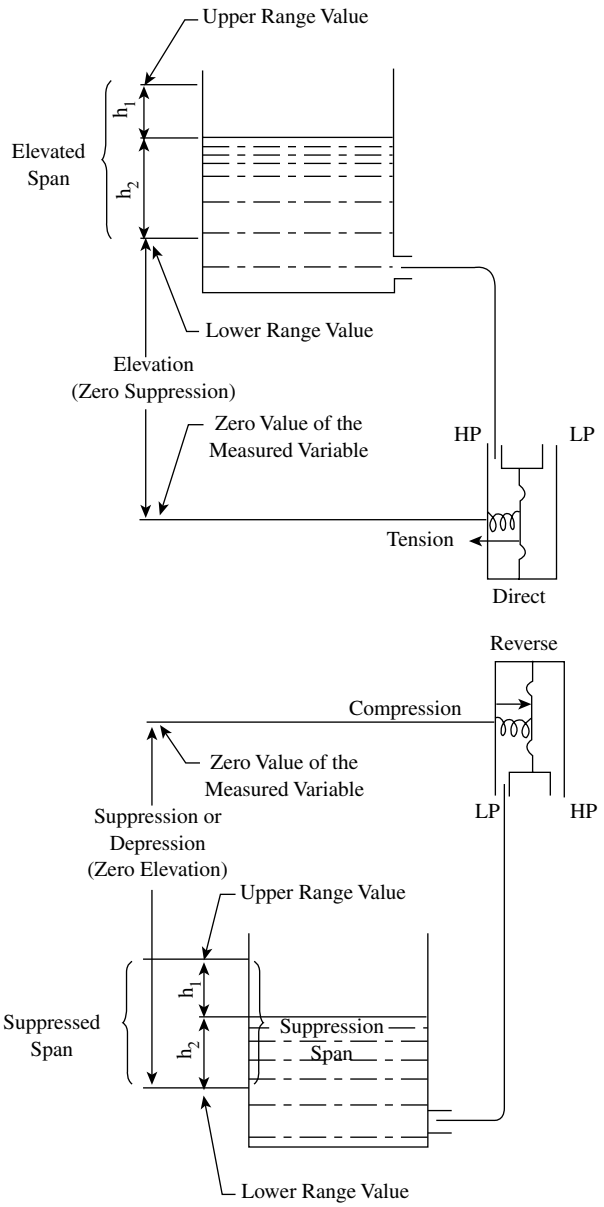


FIG. 3.6o
Illustration of elevation and depression in connection with d/p-type level measurement on atmospheric tanks.

gravity fluid. Once this figure is obtained, the repeatability of the unit can be checked periodically.

Figure 3.6q shows a wet-leg installation. Span is determined the same way as before ($X \times SG_1$). Range depression is calculated as the difference between the hydraulic head in the wet leg (ZSG_2) and the range elevation (YSG_1) desired. The difference between process and filling fluid densities must be selected such that the depression does not exceed the limit given in Table 3.6n. For example, if the desired minimum level is at the cell datum line ($Y = 0$), the difference between maximum and calibrated span is 200 in. (5 m) of water column, and the height of the wet leg is 100 in. (2.5 m), then the density of the filling liquid cannot be more than 2. The actual span setting of the cell can be anywhere below the full range.

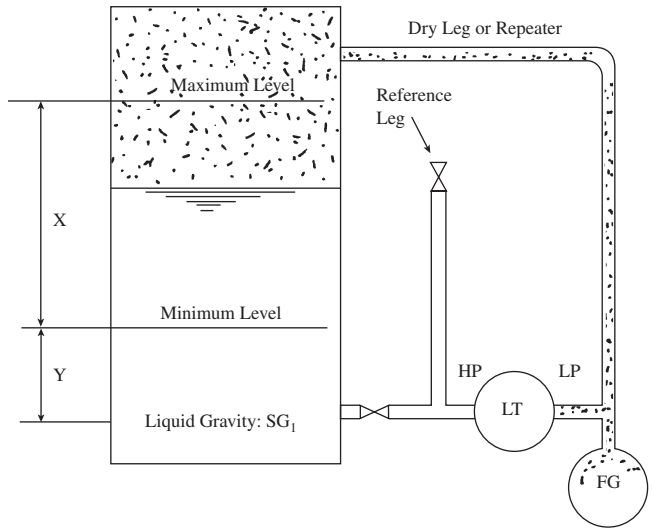


FIG. 3.6p
Illustration for range elevation; span = $X(SG_1)$ and elevation = $Y(SG_1)$.

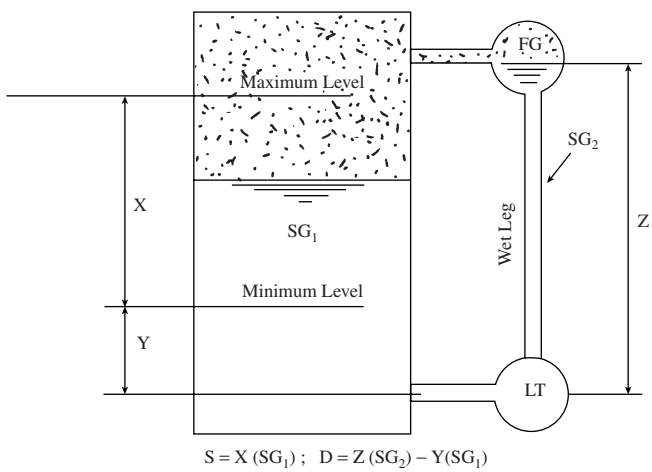


FIG. 3.6q
Illustration for range depression.

INTERFACE DETECTION

Figure 3.6r shows the settings for a liquid–liquid interface application. The span for this cell is the product of the density difference of the two liquids ($SG_2 - SG_1$) and the distance between the maximum and minimum interface levels (X). The range depression is the difference between the hydraulic head of the filling fluid ($Z \times SG_3$) and the sum of the range elevation ($Y \times SG_2$) plus the light liquid head over the range of minimum interface to overflow level ($[X + V] \times SG_1$). No depression is required if the minimum interface is at the cell datum, the height of the wet leg is the same as the maximum total level, and the filling fluid density (SG_3) is the same as the light liquid (SG_1). If the minimum interface is at the cell datum and a dry-leg system is used, then, instead of depression, the cell must

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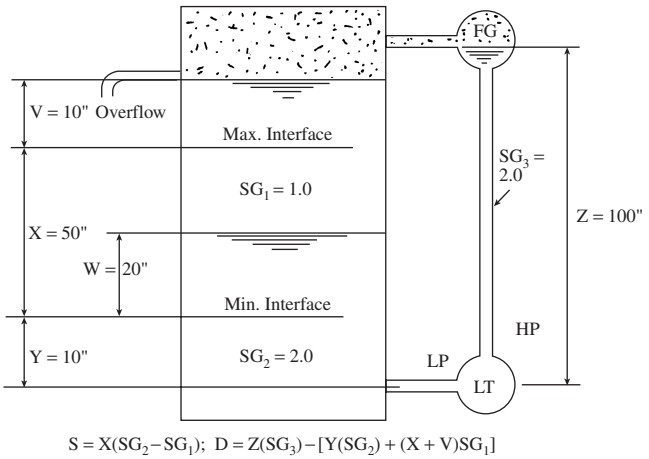


FIG. 3.6r
Span and depression settings for interface detection.

be elevated by the hydraulic head of the light liquid over the range of cell datum to overflow level ($X + V \times SG_1$).

The following calculations, which are based on the data shown in Figure 3.6r, will serve as examples.

$$\text{Wet leg hydraulic head} = Z(SG_3) = 200 \text{ in. } H_2O \quad \textbf{3.6(1)}$$

$$\text{Process side hydraulic head at minimum interface} = (V + X)SG_1 + Y(SG_2) = 80 \text{ in. } H_2O \quad \textbf{3.6(2)}$$

$$\text{Process side hydraulic head at maximum interface} = V(SG_1) + (X + Y)SG_2 = 130 \text{ in. } H_2O \quad \textbf{3.6(3)}$$

$$\text{Transmitter span} = X(SG_1 - SG_2) = 50 \text{ in. } H_2O \quad \textbf{3.6(4)}$$

$$\text{Range depression} = Z(SG_3) - [Y(SG_2) + (X + V)SG_1] = 120 \text{ in. } H_2O \quad \textbf{3.6(5)}$$

To determine the correct output signal from the transmitter at any known interface level, the proper result is simply the percentage of the transmitter span represented by the known level. If *W* is the known level, then 20 in. represents 40% of the 50-in. span. For a 3- to 15-PSI output, this will be 40% of 12 (= 4.8) plus 3 PSI for the "live zero," giving a final answer of 7.8 PSIG. Similarly, for a 4- to 20-mA output, the result will be 40% of 16 (= 6.4) plus 4 mA for the "live zero," to give a final answer of 10.4 mA.

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